Coset trialities, chamber recognition and reduced relations

Human Compression project: recovery of L122

Leemans and Stokes [1] construct coset geometries with trialities and compare their automorphism groups with those of reduced incidence graphs. We give the complete elementary interfaces needed by their finite examples. The subgroup construction and reduction are their ideas; the order and graph conventions are made explicit. A type-permuting correlation need not have order equal to the length of its type cycle.

1 The subgroup action and its exact order

Let G be a group and (H_0, H_1, H_2) an indexed subgroup family. The objects of type i are the right cosets $H_i g$; objects of different types are incident when their underlying cosets intersect. The right action of G preserves incidence. No assertion that it is the full automorphism group follows merely from this construction.

Proposition 1. *Suppose $\sigma \in \operatorname{Aut} G$ maps H_i to H_{i+1}, with indices modulo three. Then*

$$\tau(H_i g) = H_{i+1}\sigma(g)$$

is an incidence-preserving type cycle. It has order three if and only if $H_i\sigma^3(g) = H_i g$ for every i, g. In particular, $\sigma^3 = 1$ suffices.

Proof. The map is well defined because it sends every equality of cosets through a group automorphism. It is bijective, and a common coset element maps to a common coset element; applying the inverse proves the converse incidence implication. Its cube sends $H_i g$ to $H_i\sigma^3(g)$, giving the exact criterion. The type cycle rules out orders one and two. □

The same proof works for any number r of cyclically permuted types, with σ^r in place of σ^3. The requirement that the pregeometry is a geometry remains separate. Lemma 1 and Method 1 of [1] need this order qualification to identify the *induced* correlation as a triality under their initial definition. We do not infer that their stronger bare existence assertion is false solely from the omitted order argument.

2 A complete rank-three recognition criterion

Theorem 2. *Let $G = \langle r_0, r_1, r_2\rangle$, where the r_i are distinct involutions, and put $H_i = \langle r_j, r_k\rangle$ for $\{i, j, k\} = \{0, 1, 2\}$. Suppose*

$$H_i \cap H_j = \langle r_k\rangle, \qquad \bigcap_i H_i = 1, \qquad H_0 \cap H_1 H_2 = (H_0 \cap H_1)(H_0 \cap H_2). \tag{1}$$

Then the coset system is a thin, residually connected, flag-transitive geometry. Its full type-preserving automorphism group is the given right copy of G.

Proof. Every incident pair of cosets has a common element, and so is a translate of the corresponding pair of standard parabolics. Consider a triple of pairwise incident cosets. Translate the intersection of its last two cosets to the identity, leaving the triple $H_0 g, H_1, H_2$. Choose

$h_1 \in H_0 g \cap H_1$ and $h_2 \in H_0 g \cap H_2$. Then $h_1 h_2^{-1} \in H_0 \cap H_1 H_2$ and (1) writes it as uv with $u \in H_0 \cap H_1$, $v \in H_0 \cap H_2$. The element $u^{-1} h_1 = v h_2$ belongs to all three cosets. Thus every chamber has a common element and is a translate of (H_0, H_1, H_2). Every incident pair extends to such a chamber, as does every singleton. This proves the geometry and flag-transitivity assertions.

The chamber stabilizer is $\bigcap H_i = 1$, so G acts regularly on chambers. A panel missing type i has exactly $|H_j \cap H_k| = 2$ completions. The rank-two residue at H_i is the coset graph of H_i with the two subgroups $\langle r_j \rangle$ and $\langle r_k \rangle$, which generate H_i; it is connected. This last connectivity can be seen directly by writing a group element as an alternating product of the two subgroup generators and following the corresponding incident cosets. The full incidence graph is connected for the same reason, since the three H_i generate G. These are all residues of rank at least two.

The two chambers in the panel of the standard chamber missing type i are the standard chamber and its translate by r_i. The chamber adjacency graph is therefore the connected Cayley graph generated by the r_i. A type-preserving automorphism fixing one chamber fixes its unique neighbor of each type, and induction along this graph fixes every chamber and object. Any type-preserving automorphism can first be composed with an element of G to fix the standard chamber. It is then the identity. This proves the full automorphism assertion, without a graph-automorphism census. □

In the finite certificates, each side on the right of (1) has order two and they intersect trivially. Their product has four elements and is always contained in the left side. Thus computing that the left side has four elements proves equality. This is a proof premise, not a heuristic order comparison. The certificates check all three cyclic versions as well as all 64 subgroup intersections of the marked triple.

3 Lossless reduction for an actual triality

Assume that τ is an incidence-preserving correlation of order three cycling all types of a rank-three pregeometry. Let V be its type-zero objects and define the directed relation

$$A(x, y) \quad \Longleftrightarrow \quad x \text{ is incident with } \tau(y). \tag{2}$$

Loops are permitted. This is the reduced oriented graph, with the absolute vertices marked by its loops. In the graphical convention of [1], these marks are called half-edges.

Theorem 3. *Relation (2) reconstructs the entire typed pregeometry. Its automorphism group is naturally the centralizer of τ in the full type-preserving automorphism group of the pregeometry.*

Proof. Every object has a unique expression $\tau^i(x)$ with $x \in V$ and $i \in \{0, 1, 2\}$. For different types i, j, their incidence is $A(x, y)$ when $j - i = 1$ modulo three, and is $A(y, x)$ when $j - i = 2$. Indeed apply τ^{-i}; in the second case use symmetry and $\tau^3 = 1$. This proves reconstruction. An automorphism f of (2) extends by $\widehat{f}(\tau^i x) = \tau^i f(x)$. The incidence formulas prove that this extension is a type-preserving automorphism commuting with τ. Conversely such an automorphism restricts to an automorphism of (2), and commutation makes its action everywhere determined by that restriction. These operations are inverse group homomorphisms. □

For the coset action of Proposition 1, conjugation by τ on the faithful right copy of G induces σ. If Theorem 2 applies, Theorem 3 consequently identifies the full reduced automorphism group with G^σ. In particular, computing only that subgroup would not suffice without the full typed-group argument.

4 Darts, voltages and higher-order qualifications

For an order-r correlation cycling r types, use coordinates $(x, i) = \sigma^i x$. Record a directed incidence from $(x, 0)$ to (y, d) by the voltage $d \in \mathbb{Z}/r\mathbb{Z}$. Its reverse record is $(y, x, -d)$. The

original incidence graph is reconstructed by lifting it to all edges (x, i)–$(y, i + d)$. Every vertex and every *dart* orbit has size r; vertices and darts therefore divide in number by r. This proves the quotient-cover assertion in the dart convention, including its exact lift. An undirected edge orbit can instead have size $r/2$ when a half-turn interchanges its endpoints. Such an orbit is a quotient semiedge. A quotient loop with nonzero voltage is a different object: for $r = 3$ an absolute vertex lifts to a triangle, so its mark must retain that voltage-loop meaning. The single-dart semiedge convention appropriate to a polarity is not an ordinary graph-cover interpretation of this triality mark.

The order hypothesis cannot be removed by redefining the correlation on its last type, as suggested after Lemma 4 of [1]. Here are elementary counterexamples to that suggested normalization; neither changes the order-three examples of the application.

Proposition 4. *There are rank-two and rank-three geometries with cyclic type-permuting correlations for which resetting the last type map to make its cube or square the identity does not preserve incidence.*

Proof. For rank two take the cycle on $\mathbb{Z}/6\mathbb{Z}$, with the even and odd vertices as types, and $\delta(x) = x + 1$. The proposed reset agrees with δ on even vertices and with δ^{-1} on odd vertices. It sends the incident pair $(1, 2)$ to $(0, 3)$, which is not incident. The reduced directed relation on the even vertices has loops and a directed three-cycle. Discarding its orientation identifies the two opposite relations and is not the claimed lossless unlabeled reduction.

For rank three take $\mathbb{Z}/9\mathbb{Z}$, types modulo three, and incidence between vertices whose cyclic distance is one or two. Every edge lies in a consecutive triple, so this is a geometry. Each vertex residue is a path on four vertices, so it is residually connected. Rotation $\sigma(x) = x + 1$ has order nine. Resetting the last type map replaces its value on type two by σ^{-2}. It again sends the incident pair $(2, 3)$ to $(0, 4)$, which is not incident. The natural degree-three voltage lift also fails without the order condition: the reduced voltage-two record from 3 to 0 represents the edge $(3, 2)$, whereas its sheet-one lift would be $(4, 0)$, a nonedge. Thus the general cover assertion of Lemma 4 needs the same qualification, not merely its identification as a quotient. For completeness its full graph automorphism group is dihedral of order 18: among its edges those of distance one have two common neighbors, whereas those of distance two have one; hence its automorphisms preserve the underlying nine-cycle, and every dihedral symmetry is an automorphism. Every element of order three is rotation by three or six, preserving each type. Thus even the existence of an order-three type cycle does not follow for arbitrary geometries from the existence of an order-nine type cycle. This last geometry is not being asserted to be the coset system of Proposition 1; it is not a counterexample to that lemma's stronger bare existential formulation. □

For an arbitrary correlation cycling types, one may still retain every oriented labeled relation between the selected coordinate types. Those relations reconstruct incidence by applying the actual powers of the correlation. They do not justify imposing a new order or discarding the extra relations. The repaired order-r version above is the one used here. More explicitly, for arbitrary order define $A_d(x, y)$ by $x * \sigma^d(y)$, for $1 \le d < r$. If $i < j$, incidence of $\sigma^i x$ and $\sigma^j y$ is exactly $A_{j-i}(x, y)$, by applying σ^{-i}. Thus all the labeled relations are lossless without any order assumption; what fails is the proposed order-r voltage lift or the deletion of the extra data. In rank two a single directed relation suffices. Different dualities can give different reductions: on the six-cycle, rotation $x \mapsto x + 1$ has three reduced loops, whereas the polarity $x \mapsto 1 - x$ has two. This also proves the nonuniqueness qualification of source Lemma 2.

References

[1] D. Leemans and K. Stokes, *Coset geometries with trialities and their reduced incidence graphs*, Acta Math. Univ. Comenianae **88**(3) (2019), 911–916.

[2] R. Artzy, *Self-dual configurations and their Levi graphs*, Proc. Amer. Math. Soc. **7** (1956), 299–303.

[3] T. Pisanski and B. Servatius, *Configurations from a Graphical Viewpoint*, Birkhäuser, 2013. Quotient and voltage terminology; the incidence and reconstruction arguments used here are given in full.